JUEE JAHAGIRDAR

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RESEARCH INTERESTS

Interpretability & World Models: Belief state geometry in transformers, spectral signatures in LLM activations, architecture-dependent failure modes.
Dynamical Systems: Invariant set computation via RKHS variational optimization; attractor geometry
AI Safety & Threat Models: Goal stability and distributional robustness as alignment-relevant properties; inner alignment failures in sequence predictors; monitoring limits across architectures.
Bayesian Sequence Prediction: Non-stationary environments, model switching (CTS/CTW), limits of history accumulation under distributional shift.

EDUCATION

Indian Institute of Technology (IIT) Bombay Mumbai, India
B.Tech in Mechanical Engineering; CPI:7.80/10.0 *2023 – 2027*

- **JEE Advanced:** AIR 2902 (Top ∼2%).
- **Relevant Courses:** Linear Algebra, Probability, Complex Analysis, Statistical Machine Learning, Applied Data Science and ML, Advanced ML.

PUBLICATIONS & PREPRINTS

Jahagirdar, J., Hamzi, B., Owhadi, H., Kevrekidis, Y., "Hausdorff Metric-Based Kernel Flows for Learning Invariant Sets in Dynamical Systems: A Variational Approach." *Physica D: Nonlinear Phenomena.* [DOI] (2026)

RESEARCH EXPERIENCE

Variational Kernel Flows for Invariant Sets Nov 2025 – May 2026
Research Collaborator (Advisor: Dr. Boumediene Hamzi) *Caltech & Imperial College London (Remote)*

- Implemented the full benchmark pipeline for a variational RKHS framework learning general invariant sets (including saddle-type and non-attracting structures) directly from time-series data; evaluated across 130 in-scope chaotic systems from `dysts` spanning $d = 3$ to $d = 6$.
- Designed the bilevel optimization protocol (inner L-BFGS on invariant-set energy, outer kernel update via cross-validation), train/test partitioning, and failure mode taxonomy categorizing 37 out-of-scope systems by structural cause (DDEs, non-autonomous, hyperchaotic, discontinuous RHS).
- Results: 98/130 successful reconstructions (HD < 5.0), median HD = 0.323; performance degrades gracefully from median HD = 0.079 at $d = 3$ to 0.473 at $d \geq 6$.

Belief State Geometry in Non-Ergodic Transformers Mar 2026
Independent Researcher

- Pre-registered predictions; confirmed residual stream linearly encodes analytical belief states, four process-type clusters separating as context grows – consistent with all pre-registered predictions.
- Discovered CW-trained models map unseen CCW sequences onto the **same geometric manifold** (r=0.92–0.96, inter/intra-cluster ratio 1.005): the non-ergodic boundary is invisible to the learned representation.
- Connected belief crystallization to CTS model-switching: a model trained on one ergodic component behaves like CTW with one model, no mechanism to adapt across a boundary it never crossed.

Temporal Dynamics & Spectral Signatures in LLM Activations Dec 2025 – Jan 2026
Independent Researcher

- **Discovered architecture-specific failure signatures**: spectral entropy boosted hallucination detection in Gemma-2-2B but not Qwen2.5, revealing limits of cross-architecture monitoring tools.
- Established a task-aware detection framework: spectral features excel for subtle hallucinations; static features remain superior for geometrically-obvious failure modes.

CTS in Monte-Carlo AIXI Jul 2025 – Nov 2025
Implementer (Collaborator: Cole Wyeth), University of Waterloo *Remote*

- Implemented a **Factored CTS Policy** for a Monte-Carlo AIXI agent, decoupling action-probability modeling from environment dynamics; variable-length context lookups handle sequential dependencies between action bits.
- Integrated into MCTS planner for non-stationary GridWorld; CTS recovers from rule changes that cause CTW-based agents to fail permanently.

LEADERSHIP & SERVICE

DAMP Mentor May 2025 – April 2026
Department Academic Mentorship Program *IIT Bombay*

- Selected from 100+ applicants for a 50-member cohort to mentor sophomores; coordinated early intervention strategies for at-risk students.